

Project Portfolio

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Introduction

My name is Vatsal Gohel. I currently serve as Director of Institutional Research at Clinton Community College, where I lead data strategy, accreditation, and institutional planning. I hold a Master of Science in Data Analytics (STEM) from Clark University and bring extensive experience extracting actionable insights from complex datasets to drive strategic decision-making and optimize organizational processes.

Professional Summary

I am a results-driven institutional research professional with a strong technical analytics background and the ability to translate complex data into clear, evidence-based insights for executive leadership and external stakeholders. In my current role at Clinton Community College, I lead all institutional research, strategic planning, and academic assessment functions, serve as Accreditation Liaison Officer (ALO) for MSCHE, and oversee accreditation reporting and compliance. I represent the college at SUNY and professional IR organizations and conferences.

Prior to this, I served as a Data Analyst at Clinton Community College — leading all NYSED, SUNY, IPEDS, and MSCHE data submissions and developing a KPI tracking dashboard for the college's Strategic Planning initiative. I have additionally worked as a Data Analyst at Pinnacle Software Solutions and DynaTech Systems Inc., building scalable Power BI dashboards and SQL-based reporting solutions. At Clark University I served as Teaching Assistant supporting 60+ graduate students. At Numerator, I progressed from Data Quality Associate to Research Analyst and was recognized as Employee of the Year for high-impact delivery across global CPG client portfolios.

Tools Used and Known

I am proficient in a wide range of tools spanning analytics, visualization, and institutional research:

- Visualization & BI: Power BI, Tableau — interactive dashboards, KPI tracking, institutional reporting
- Programming: Python (Pandas, NumPy, Seaborn, Plotly, Matplotlib), R — statistical analysis, machine learning, ETL scripting
- Data Management: SQL / SQL Server, Advanced Excel, ETL processes, data warehousing
- Institutional Research: IPEDS, NYSED, SUNY, MSCHE reporting; accreditation compliance; program review; assessment
- Cloud & Infrastructure: Azure, AWS; SSRS (SQL Server Reporting Services)
- Certifications: Google Analytics, Power BI for Data Analysts (Microsoft), BCG Data Science Job Simulation, AWS Cloud Practitioner CLF-C02, Anaconda Python for Data Science, SQL for Data Analysis, Excel: Advanced Formulas & Functions

12. Higher Education Enrollment Dashboard

Project Title

Automated Enrollment Intelligence Dashboard for Clinton Community College

Problem Statement

Clinton Community College relied on its registrar to manually pull and distribute enrollment reports daily. This created critical operational gaps: reports were not generated on weekends, holidays, or during the registrar's absence — yet student enrollment continued uninterrupted. When two academic terms overlapped (e.g., Summer and Fall registration running simultaneously), combining dual-term data into a single coherent report presented a further analytical challenge. Academic leaders had no always-available, centralised view of enrollment health.

Project Description

Objective: To design and deliver an interactive enrollment intelligence dashboard that consolidates multi-year enrollment data, eliminates manual reporting gaps, and establishes the architecture for a fully automated, real-time refresh pipeline.

Scope:

- Consolidate registrar-generated reports from AY 2019–20 through AY 2025–26 into a single structured dataset.
- Build a four-page interactive Power BI dashboard presenting Headcount, FTE, and Credits across enrollment status, enrollment type, dual enrollment, academic year, and term dimensions.
- Automate data consolidation via a Python ETL script that regenerates the master workbook whenever new reports are deposited.
- Implement a dual-folder file management strategy (Main + Version folders) to maintain stable Power BI data-source links across refresh cycles.
- Design architecture to support a future fully-automated Phase 2 pipeline via SSRS and scheduled jobs.
- Create a synthetic dummy dataset replicating real enrollment patterns for open GitHub publication without exposing institutional data.

Deliverables:

- Python ETL consolidation script.
- Master Excel workbook (FTE Combined Output) — the Power BI data source.
- Four-page interactive Power BI dashboard: Home, Trends, Deep Dive, and Definitions.
- Dual-folder file management structure (Main + Version).
- Synthetic dummy dataset for open-source sharing.

Methodology

Tools & Technologies: Power BI (dashboard), Python (ETL / consolidation), SQL Server / SSRS (Phase 2 target), Excel (master workbook), GitHub (version control)

Architecture — Data Flow:

- Registrar uploads SSRS-generated enrollment reports to a designated shared folder.
- Python consolidation script reads all report files, standardises column names, data types, and date formats, then appends them into a single master Excel workbook. Re-running the script regenerates the workbook with the latest data.

- Master workbook saved under a constant filename in the Main folder — the Power BI data-source link never needs updating. Version folder stores timestamped copies for audit and rollback.
- Power BI connects to the Main folder workbook. Phase 1: manually triggered refresh. Phase 2: fully scheduled automatic refresh.

Dashboard Pages:

- Home: Executive KPIs (Headcount, FTE, Credits) for the current term pair with YoY variance indicators (VS LY Same Date — pacing; VS LY Final Report — benchmark). Trend lines from AY 2019–20 to present. Interactive calendar for point-in-time historical lookups.
- Trends: Daily pacing view active during registration periods. Compares current enrollment against the same calendar date in the prior year and against the prior year's final figures.
- Deep Dive: Granular breakdowns by Enrollment Status (Continuing, New First-Time, New Transfer, New CAP, Non-Matriculated, Cross-Registered, Continuing Non-Matric, Continuing P-Tech) and Enrollment Type (Full-Time vs. Part-Time). Variance bar charts and historical academic-year trend comparisons.
- Definitions: Built-in reference page explaining every metric, filter, and enrollment category — enabling self-service use without contacting the data team.

Filters & Interactivity:

- Term Display slicer: isolate any semester or CTRL-click to select multiple terms simultaneously.
- Dual Enrollment toggle: CAP vs. Non-CAP.
- Enrollment Type toggle: Full-Time vs. Part-Time.
- Interactive calendar: clicking a specific date forces the dashboard to show enrollment as it stood on that exact date.
- Reset button: clears all active filters and returns dashboard to its default unfiltered state.

Dummy Dataset Note:

No actual institutional enrollment records were used in this version. A synthetic dataset was created to replicate the statistical patterns, proportions, and seasonal shapes of real enrollment data, allowing the full dashboard to be demonstrated publicly on GitHub without any privacy concerns.

Role and Responsibilities

Role: Project Lead & Co-Developer

- Conceptualised the dashboard initiative; defined architecture, scope, and data flow design.
- Directed and coordinated the intern (data analyst) who carried out the bulk of the technical build.
- Aligned the intern's implementation with the intended analytical vision — resolving differences in overlapping-term handling and enrollment status categorisation.
- Reviewed, validated, and refined all dashboard pages and ETL logic.
- Designed the dual-folder file management strategy and the dummy dataset approach.
- Planned and documented the Phase 2 automation roadmap.

Challenges and Solutions

Challenge	Solution
Registrar availability gaps (weekends, holidays, leave)	Replaced manual distribution with a self-service Power BI dashboard accessible at any time, removing single-person dependency.
Overlapping term reports (Summer & Fall active simultaneously)	Dual-term ETL handling; combined totals shown by default; Term slicer allows single-term isolation.

Inconsistent report formats across years (column names, date formats)	Python ETL normalises column names, data types, and date formats before writing to the master workbook.
Intern's implementation diverging from analytical intent	Regular design reviews, written requirements, and iterative feedback cycles kept implementation aligned.
Stable Power BI data-source path across refresh cycles	Constant-filename Main folder for Power BI; Version folder for timestamped audit copies.
Phase 1 requires manual steps (Python run + Power BI refresh)	Deliberate Phase 1 constraint; architecture is Phase-2-ready for full SSRS → Python → Power BI automation.
Data privacy for GitHub/open-source publication	Synthetic dummy dataset created — replicates statistical patterns without any real student or institutional records.

Results and Impact

Outcome:

- Fully functional four-page Power BI dashboard covering AY 2019–20 through AY 2025–26.
- Python ETL script consolidating any number of registrar reports into the master workbook in a single run.
- Dashboard accessible independent of registrar availability — reporting gaps on weekends and holidays eliminated.
- Dummy dataset enabling clean open-source publication on GitHub.

Impact:

- Academic decision-makers gain a self-service view of enrollment health — Headcount, FTE, and Credits — across every term and status category.
- YoY pacing metrics enable early identification of enrollment trends and timely institutional intervention.
- Architecture provides a clear, low-friction path to full Phase 2 automation with zero manual steps.

Phase 2 Roadmap (Planned)

Phase 2 will eliminate the two manual Phase 1 steps (Python run + Power BI refresh trigger):

- SSRS configured to automatically export new enrollment reports to the shared folder on a schedule.
- Scheduled Python job (Windows Task Scheduler or Azure Automation) consolidates files automatically whenever new reports arrive.
- Power BI scheduled refresh picks up the updated master workbook and refreshes all visuals automatically.
- End result: fully automated, zero-touch enrollment intelligence pipeline from source system to published dashboard.

Conclusion and Learnings

This project demonstrated the value of combining lightweight ETL scripting with a modern BI platform to eliminate manual reporting dependencies in a higher-education context. The dual-folder strategy, dummy dataset approach, and phase-based automation roadmap are reusable patterns applicable to similar institutional reporting challenges.

Key Learnings:

- Stakeholder and collaborator alignment is as important as technical execution — differing interpretations of enrollment metrics required explicit documentation and iterative review.
- Designing for automation from the start — even in a manual Phase 1 — significantly reduces rework in subsequent phases.
- Synthetic dummy datasets are a practical solution for demonstrating analytics work on sensitive institutional data in public portfolios.
- Overlapping-term complexity in higher-education enrollment requires deliberate ETL and UX design; it cannot be treated as a simple additive sum.

Link: [Higher Education Enrollment Dashboard](#)